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Using combinations of principal component scores from different spectral ranges in near-infrared region to improve discrimination for samples of complex composition

Sanguk Lee [a], Hoeil Chung [a,*], Hangseok Choi [b], Kyungjoon Cha [c]

[a] Analytical Spectroscopy Lab, Department of Chemistry, Hanyang University, Seoul, 133-791, Republic of Korea
[b] Center for Genome Research, Samsung Biomedical Research Institute, Gangnam-gu, Seoul, 135-710, Republic of Korea
[c] Department of Mathematics, Hanyang University, Seoul, 133-791, Republic of Korea

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ABSTRACT

Principal component analysis (PCA) is widely used as an exploratory data analysis tool in the field of vibrational spectroscopy, particularly near-infrared (NIR) spectroscopy. PCA represents original spectral data containing large variables into a few feature-containing variables, or scores. Although multiple spectral ranges can be simultaneously used for PCA, only one series of scores generated by merging the selected spectral ranges is generally used for qualitative analysis. Alternatively, the combined use of an independent series of scores generated from separate spectral ranges has not been exploited.

The aim of this study is to evaluate the use of PCA to discriminate between two geographical origins of sesame samples, when scores independently generated from separate spectral ranges are optimally combined. An accurate and rapid analytical method to determine the origin is essentially required for the correct value estimation and proper production distribution. Sesame is chosen in this study because it is difficult to visually discriminate the geographical origins and its composition is highly complex. For this purpose, we collected diffuse reflectance near-infrared (NIR) spectroscopic data from geographically diverse sesame samples over a period of eight years. The discrimination error obtained by applying linear discriminant analysis (LDA) was improved when separate scores from two spectral ranges were optimally combined, compared to the discrimination errors obtained when scores from singly merged two spectral ranges were used.

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1. Introduction

Principal component analysis (PCA) is an exploratory data analysis tool widely utilized in the field of vibrational spectroscopy. It is an efficient data reduction method capable of representing a large amount of spectral data with a few variables (scores) while retaining important analytical information present in the original spectra [1,2]. When samples are discriminated using spectroscopic methods, PCA is frequently combined with other discrimination methods such as linear discriminant analysis (LDA). LDA is one of most frequently utilized methods for discrimination, especially in the field of vibrational spectroscopy. It is a multivariate method used to classify objects into one of two or more groups by maximizing the ratio of between-class variance to within-class variance. Numerous detailed descriptions of LDA can be found elsewhere [3–5].

When PCA is used for the spectroscopic discrimination of samples, it is always desirable to maximize the separation of corresponding scores by optimizing necessary parameters such as spectral range before performing LDA. When samples with greatly overlapping spectral features due to the complex composition of natural products are analyzed, it is difficult to determine the optimal spectral range by simply examining a specific spectral feature (or range) of a given component (or a group of components) that functions as a discrimination marker. From a practical standpoint, an optimal range should be found empirically and then validation of the data would follow. To select the optimal spectral range, several approaches, such as moving window PCA (MWPCA), have been investigated previously [6–8]. In MWPCA, large numbers of spectral ranges are generated by changing the size and position of a spectral window across a spectral range and then identifying the best range by examining the degree of differentiation (or discrimination error) at each spectral range. The use of MWPCA has been shown to improve differentiation of mastitic and healthy udder quarters in cows using visible and NIR spectroscopy [8]. Additionally, the combined use of different spectral ranges has been demonstrated to improve the performance of partial least squares (PLS) regression [9–12]. Although these studies were designed for quantitative analysis, the same strategy may be equally applicable if used to maximize the separation of groups and thereby improve discrimination performance.

Although the methods listed above may be successful for identifying the best spectral ranges for optimal discrimination, their

* Corresponding author. Tel.: +82 2 2220 0937; fax: +82 2 2299 0762.
E-mail address: hoeil@hanyang.ac.kr (H. Chung).

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strategies are based on the use of a series of scores generated from only a single range or a combination of multiple ranges found by an optimization procedure. The possible use of scores independently generated from separate spectral ranges that are optimally combined to improve the discrimination of groups, however, has not been investigated. For example, when two different ranges are used for PCA, a single series of scores can be obtained by merging two ranges (as described above), or two separate series of scores can be acquired from each spectral range.

The purpose of this study is to investigate whether it is possible to improve the discrimination of groups by optimally combining two separate series of scores, referred to as Independent Score Combination PCA (ISC-PCA). For purposes of comparison, a graphical depiction of ISC-PCA is presented in Fig. 1(a) and the conventional procedure of PCA for discrimination is shown in Fig. 1(b). The shaded blocks correspond to two spectral ranges used for analysis. When two spectral ranges are used in a conventional PCA scheme, one series of scores generated from a single merged range is utilized for discrimination (Fig. 1(b)). In the ISC-PCA scheme, two series of scores generated from two separate ranges are separately utilized for discrimination as shown in Fig. 1(a). In both cases, LDA was used to determine the corresponding discrimination error. The combined use of independent factors (eigenvectors) from separate spectral ranges may highlight minute spectral differences more effectively than the use of factors derived from the combined spectral range.

To evaluate the utility of ISC-PCA for group discrimination, we have attempted to differentiate between two geographical origins of sesame with the use of NIR spectroscopy. Sesame is a typical agricultural product. Its composition is highly complex, and the resulting spectra substantially overlap without recognizable distinct

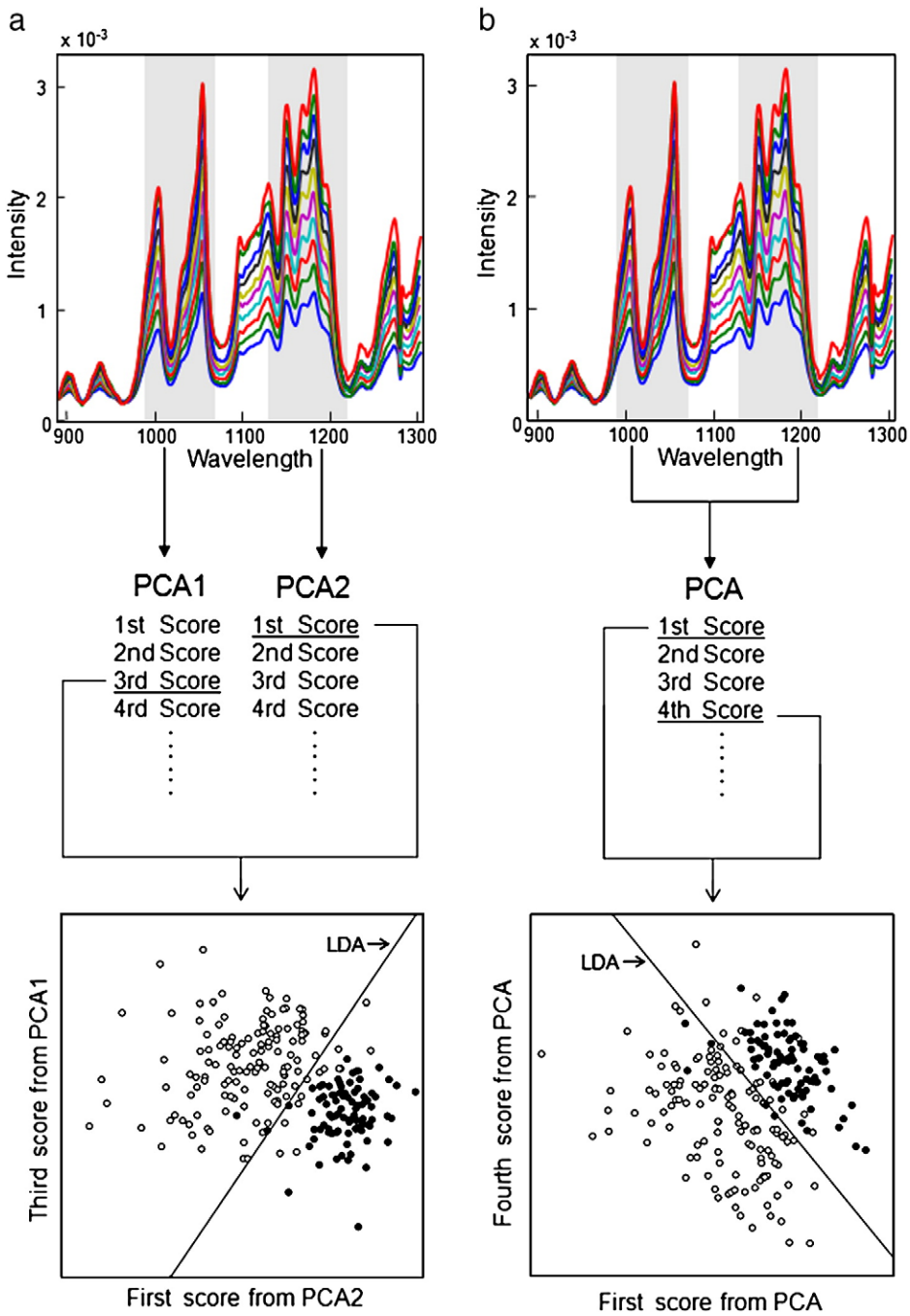

Fig. 1. Graphical descriptions of ISC-PCA (a) and conventional PCA for discrimination. The shaded blocks correspond to two spectral ranges used for analysis.

features from specific components. Because of this, the effective recognition of subtle spectral differences by which the geographical origins of sesame could be distinguished would be highly desirable. The resulting discrimination accuracies from both schemes present in Fig. 1 have been compared to each other.

2. Experimental procedures

2.1. Sample and NIR spectral collection

A total of 1122 sesame samples (432 imported and 690 domestic Korean samples) were obtained from the National Agricultural Products Quality Management Service (NAQS) in Seoul, Korea. All samples were collected over a period of more than eight years (from 1997 to 2005) to account for wider compositional variations from seasonal changes and cultivation areas. Imported products came mostly from diverse locales in China. The raw sesame samples were ground into powder. The final samples for NIR spectral collection were fine powders prepared by passing the ground powder through a 20-mesh sieve. The preparation of fine powder for diffuse reflectance NIR spectral collection was designed to minimize possible background variation resulting from variation in sample morphology and packing. The NIR spectra were collected immediately following sample preparation.

NIR diffuse reflectance spectra were collected over the spectral region from 1100–2500 nm with a Foss NIRSystems Model 6500 spectrometer equipped with a quartz halogen lamp and PbS detector. The spectra were acquired by gently packing fine powder samples in circular sample cups (38 mm in diameter and 10 mm in thickness) with quartz windows. Each sample spectrum was obtained by averaging 32 scans. All spectra were recorded as log 1/R with respect to a ceramic reference standard at room temperature.

All algorithms were developed using Matlab Version 7.0 (The MathWorks Inc., MA, USA). PCA was conducted in Matlab and all preprocessing methods, including MSC (multiplicative scatter correction) and ISC-PCA, were also accomplished using Matlab.

3. Results and discussion

3.1. Examination of NIR spectral features

Sesame is an agricultural product constituting complex mixtures of various organic compounds and therefore, the NIR spectra of sesame samples are highly complicated and lack dominant features of specific components that could be employed as discrimination index compounds. Fig. 2(a) shows the averaged NIR spectra of imported and domestic (Korean) sesame. The spectra shown were preprocessed using multiplicative scatter correction (MSC) to reduce underlying background variation from scattering. As expected, the overall spectral features are broad and similar to each other throughout the whole spectral range (1100–2500 nm). For a more detailed comparison, the spectral features in the 1100–1700 and 1700–2500 nm ranges are presented separately in Fig. 2(b) and (c), respectively. Minute spectral differences in band shape and intensity are observed between the imported and domestic sesame in both spectral ranges. The spectral differences are larger in the 1700–2500 nm range because both the penetration depth of NIR radiation into a solid sample for diffuse reflectance measurements and the molar absorptivity of a sample are larger in the longer wavelength region. It is very hard to rationalize the spectral differences in the NIR spectra based on botanical origin. However, it is believed that the spectral variations arise from the differences in metabolites yielded by the different metabolic pathways between the imported and domestic sesame.

Under this situation, when distinct spectral bands are not readily available, the use of a multivariate method based on a statistical approach would be the proper choice for the discrimination of two geographical origins of sesame. As discussed, the spectral differences between the two groups shown in Fig. 2 are not significant. However, the optimal selection of a spectral range as well as the proper use of an adopted multivariate method is both critically important to achieve the best discrimination performance.

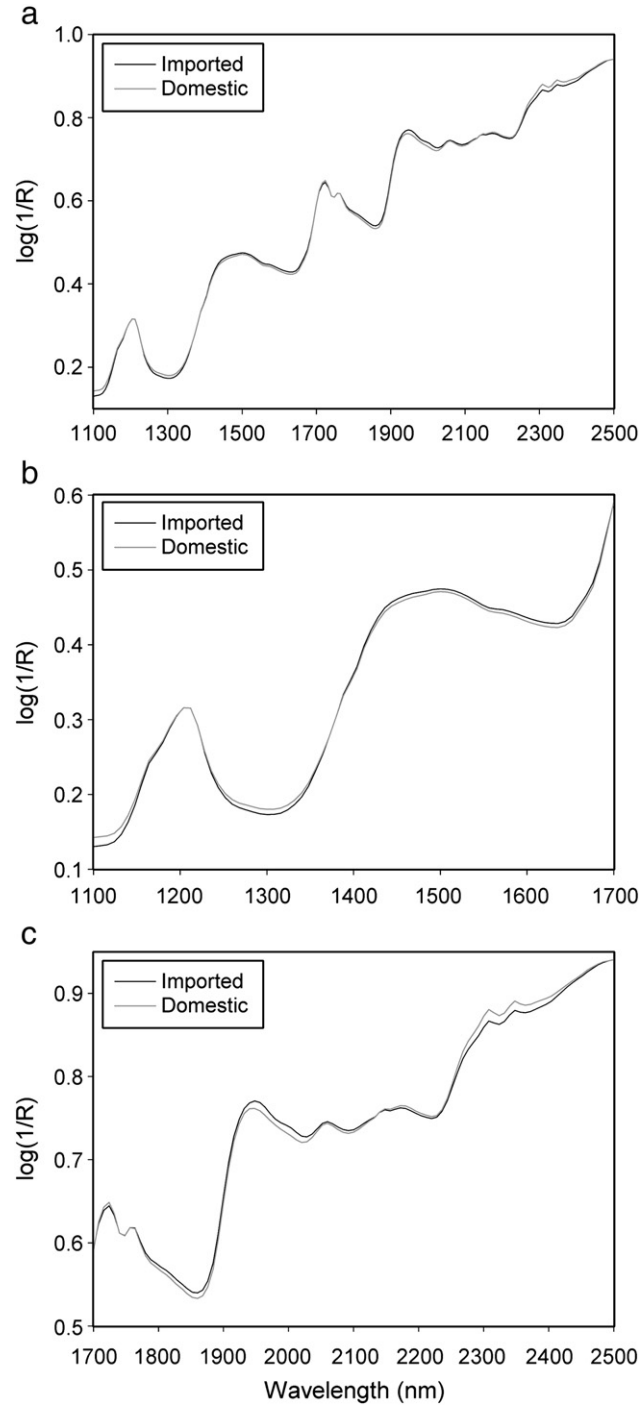

Fig. 2. Averaged NIR spectra of imported and domestic (Korean) sesame in the 1100–2500 nm range (a). The 1100–1700 (b) and 1700–2500 nm ranges (c) are highlighted for the more detail comparison.

3.2. Discrimination of two sesame groups using whole spectral range

To statistically evaluate the performance of discrimination into two different geographical origins of sesame, we have divided the entire dataset (total 1122 samples, 432 imported and 690 domestic

samples) into calibration and validation sets. A total of 346 imported and 552 domestic samples were assigned to the calibration set, and 86 imported and 138 domestic samples were assigned to the validation set. The division was intended to assign 20% of the total samples to the validation set. The calibration set was used to develop a discrimination model, and the validation set was used to evaluate the discrimination performance of a developed model.

To develop a discrimination model, the calibration set was further divided into a new calibration set (75% of the original calibration set) and a new validation set (the other 25% of the original calibration set) for cross-validation. PCA was performed in a given spectral range to generate scores using the new calibration set. Using a given combination of two scores, discrimination error was calculated using LDA. Then, using a developed model, the sample in the new validation set was predicted and the resulting discrimination was calculated. The procedure described above was repeated 200 times by randomly assigning the samples into the new calibration and validation sets. This cross-validation procedure imposed statistical significance in developing a discrimination model. Large numbers of two-score combinations derived from the generated scores are possible in a given range; therefore, the procedure described above was performed for all possible two-score combinations. When different spectral ranges were evaluated for discrimination, the entire procedure described above was repeated for each spectral range. Finally an optimal two-score combination and the corresponding optimal spectral range yielding the lowest cross-validated discrimination error in the prediction were determined.

Then, a discrimination model was developed, again using the original calibration set within the optimal score combination and the spectral range. With the use of the developed discrimination model, the final discrimination error was calculated by predicting samples in the original validation set. A combination of only two scores was used because no further significant improvement in discrimination was achieved by adopting more scores (dimensions). Additionally, it was easy to visualize the performance of differentiation in the domain of two-dimensional score distribution.

By following the calibration procedure described above, the discrimination was performed using the whole spectral range (1100–2500 nm). First, we used six scores to find an optimal score combination because the use of six factors described more than 99% of the total variation present in the dataset. Because six scores are used and only a single spectral range is evaluated, there are 15 combinations ($_6C_2$) possible when selecting two out of six scores. Table 1 presents the best combination of two scores and the resulting discrimination error when the whole spectral range is used. The first and fourth scores are combined. Fig. 3(a) shows the corresponding two-dimensional score scatter plot including the boundary line determined by LDA. Closed and open circles correspond to imported and domestic sesames, respectively. Two groups are visually identifiable; however, there is also considerable overlap between the two groups. Any new strategy that can improve the degree of separation between the two score groups would be desirable.

Table 1
Optimized spectral range, optimum score combination and discrimination errors resulting from the use of four different methods of applying PCA to discriminate sesames from two different geographical origins.

Optimization method	Spectral range optimized	Scores used	Discrimination error
Full range PCA	1100–2500 nm	1, 4	13.4%
Single range optimization	1700–1800 nm	2, 5	11.6%
Dual-range optimization	1700–1800 nm and 2300–2400 nm	1, 3	10.3%
ISC-PCA	1700–1800 nm 2400–2450 nm	2 3	6.7%

3.3. Range optimization to improve discrimination

To improve discrimination between imported and domestic sesames, the selection of an optimal spectral range would be advantageous rather than using the whole spectral range, especially when necessary information for discrimination is not highly characterized throughout the entire spectral range. We have evaluated three different approaches of optimizing the spectral range for discrimination.

First, an optimal spectral range was found by using the strategy of moving window PCA (MWPCA). The main strategy of MWPCA is to perform PCA at many different spectral ranges generated by changing the size and location of the spectral window, thus determining the spectral range that yields a minimal discrimination error. Through this process, an optimized single range is found. The window size was varied, and each window was shifted across the spectral range to cover the entire spectral range. The window size (number of spectral points) was varied from 50 to 1400 nm at increments of 50 nm. In a given window size, the window was moved from low to high wavelength in 50 nm steps until the whole spectral range (1100–2500 nm) was covered. A total of 756 spectral ranges comprising different window sizes and different spectral ranges were evaluated. Using this method across the whole spectral range, six scores were generated in each spectral range, and the resulting LDA discrimination error was calculated from the combinations of two scores.

Second, using the same approach, we found the two best combined ranges for PCA. Out of the spectral ranges generated by the moving window described above (total 756 windows), PCA was performed by merging two spectral ranges, and an optimized dual-range providing the best discrimination error (referred to as the optimized dual-range) was found. In each merged spectral range, the first six scores were evaluated for the combination.

Third, as described in the introduction, the combination of independent scores from two separate spectral ranges (ISC-PCA) was investigated. For each range, the first six scores were evaluated.

Table 1 summarizes the discrimination errors obtained from each method. Fig. 3(b), (c) and (d) shows the corresponding two-dimensional score scatter plots including the corresponding boundary lines determined by LDA for single range optimization, dual-range optimization and ISC-PCA, respectively. When comparing the results using the full and optimized single range, the discrimination error is improved when the optimized single range is used. Additionally, the optimized single range (1700–1800 nm) is fairly narrow compared to the full range. Although spectral information related to the differentiation of two groups could be present throughout the entire spectral range, the more descriptive and selective information for the differentiation appears to be localized at the 1700–1800 nm range.

Comparison of the results from the optimized single and dual-range trials reveals that the optimized single range is included in the optimized dual-range. The additional merging of the 2300–2400 nm range helps to further improve the discrimination error up to 10.3%. The improvement in the discrimination error using the optimized dual-range is marginal compared to that using the optimized single range. This indicates that the optimized single range provides most of the necessary information for the differentiation. The merging of an extra range would provide additional supplementary information rather than useful new information that could be used to substantially improve the discrimination.

The discrimination error that resulted from ISC-PCA is better than that derived from the optimized dual-range. The combination of the second score from the 1700–1800 nm range and the third score from the 2400–2450 nm range provides the best discrimination error of 6.7%. By comparison, the discrimination error from the optimized

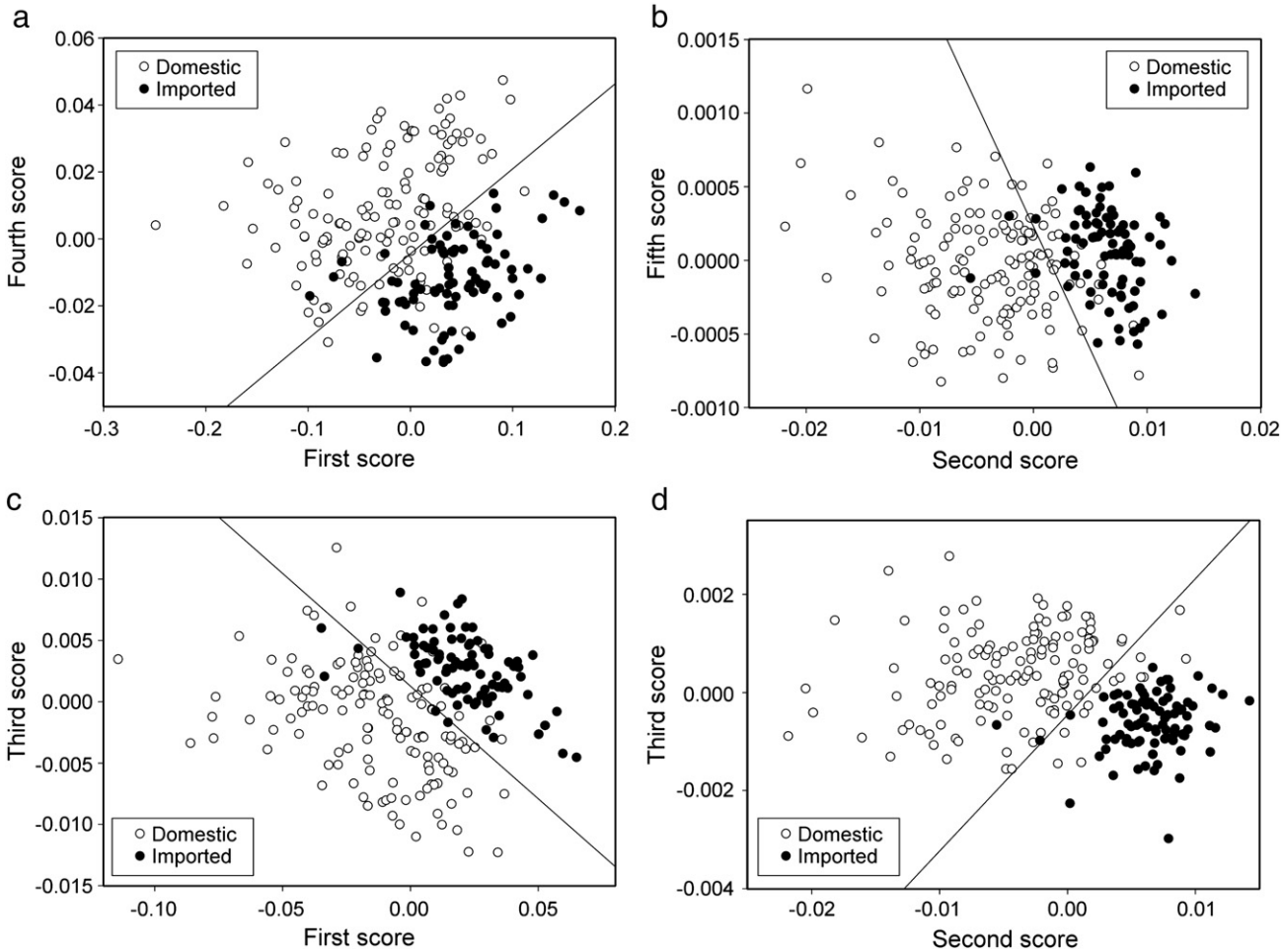

Fig. 3. Two-dimensional score scatter plots including the corresponding boundary lines determined by LDA for full range PCA (a), single range optimization (b), dual-range optimization (c) and ISC-PCA (d).

dual-range is 10.3%. The selected ranges for ISC-PCA and dual-range optimization are similar. The 1700–1800 nm range is also included for both cases, and the selected second ranges are quite close to each other. This indicates that both methods analogously utilize the spectral ranges where the most selective information for effective differentiation is available. However, the combination of two independent scores from two separate ranges leads to improved discrimination.

Fig. 4 shows the loadings used for dual-range optimization and ISC-PCA for the discrimination of sesames from two different geographical locales. The first and third loadings from the merged range of 1700–1800 nm and 2300–2400 nm are displayed for the dual-range optimization (black and dashed black lines, respectively). The second loading from the 1700–1800 nm range and the third loading from the 2400–2450 nm range are shown for ISC-PCA (gray lines). Although the 1700–1800 nm range is used for both cases, the weights of loadings of each are quite dissimilar. This indicates that the spectral descriptions of each method for the discrimination are quite different, even when the same spectral range is used.

To explain the better performance of ISC-PCA over dual-range optimization, we can conceive of a situation in which two bands are available for discrimination in a given range and the magnitudes of spectral variation of each band are different. If PCA is performed using this range, the band of larger spectral variation would be more weighted when compared to the band of smaller spectral variation during the construction of loading. Therefore, the information from the band of smaller variation could be relatively less elevated

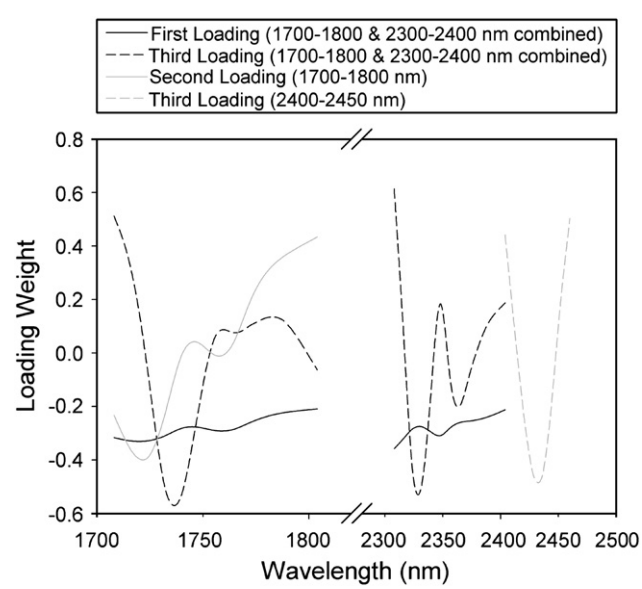

Fig. 4. Loadings used for dual-range optimization and ISC-PCA. The first and third loadings from the merged range of 1700–1800 nm and 2300–2400 nm are displayed (black lines dashed black lines, respectively). The second loading from the 1700–1800 nm range and the third loading from the 2400–2450 nm range are shown for ISC-PCA (gray lines).

although it might contain a selective feature for discrimination. On the other hand, in ISC-PCA, both bands are equally weighted and highlighted by performing PCA in a separate range. Therefore, the combination of scores from the more highlighted feature is the origin of the improvement.

4. Conclusions

Overall, the use of ISC-PCA resulted in improved discrimination of two groups of sesame samples that were distinguished by geographic origin. The improvement was based largely on the combination of more descriptive scores generated from separate spectral ranges. To ensure the general and practical applicability of ISC-PCA for discrimination, we used samples of highly complex spectral features and a long-term spectroscopic dataset to take into account possible compositional variation. If samples with simple spectral features are used for differentiation, all the methods examined in this study would work successfully and the corresponding discrimination performances would be comparable to one another. Future studies will evaluate ISC-PCA for the discrimination of more than two groups by combining ISC-PCA with advanced boundary decision methods such as support vector machines (SVM) [13]. In addition, the ISC-PCA method will be further evaluated for use in other agricultural samples using different vibrational spectroscopic methods, such as Raman spectroscopy.

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